Practical IV

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Aim: Create a Microsoft Power BI Dashboard

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1. Theory (Visualization Tools Used + Insights / Findings)

Visualization Tools Used

The COVID-19 dashboard was developed using **Business Intelligence visualization techniques** (commonly implemented in tools like Power BI / Tableau / Excel BI). Multiple visual components were integrated to provide interactive and comparative analysis of pandemic data across countries and time periods.

1. KPI Cards

- Display key aggregated metrics:
    - **Sum of Deaths**
    - **Sum of Recovered**
    - **Sum of Active Cases**
- Help users quickly understand the overall pandemic severity.
- Example: India shows ~1M deaths, 24M recovered, 16M active.

2. Map Visualization (Geo Map)

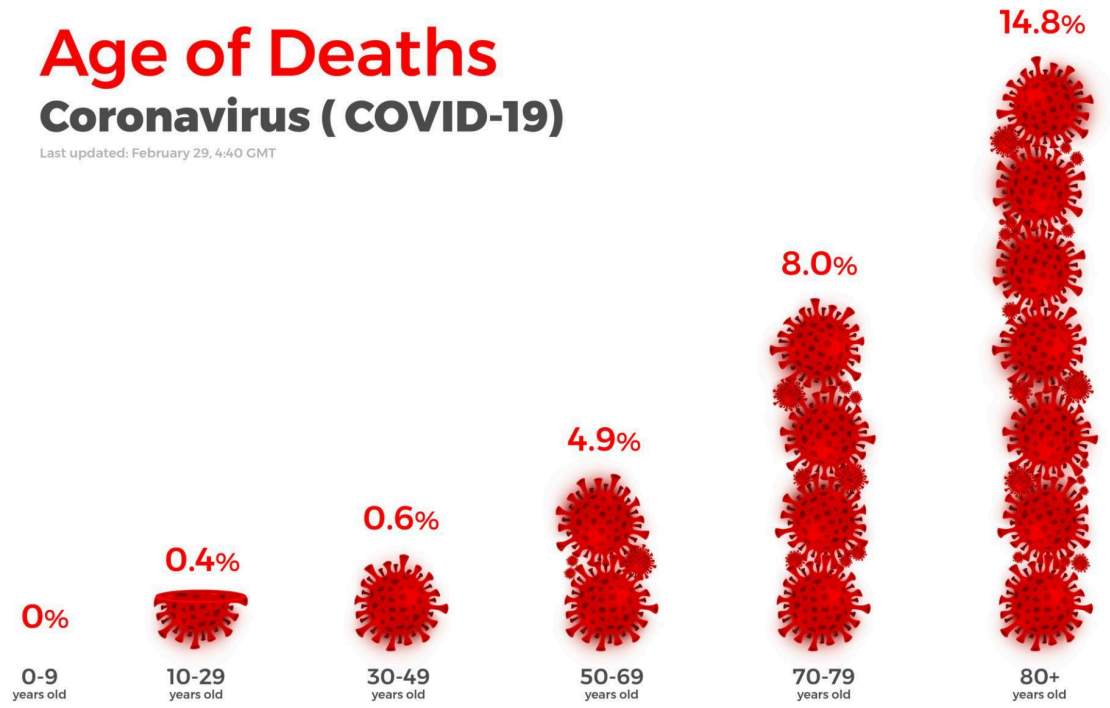
Displays **continent/country-wise spread** of COVID-19.

Enables geographic comparison.

Highlights hotspots visually.

Helps identify regional impact patterns.

3. Bar Chart – Deaths by Country/Region



- Compares total deaths across selected countries.
- Example insights:
    - India has significantly higher deaths than Cabo Verde and New Zealand.
    - Smaller nations show minimal fatality counts.

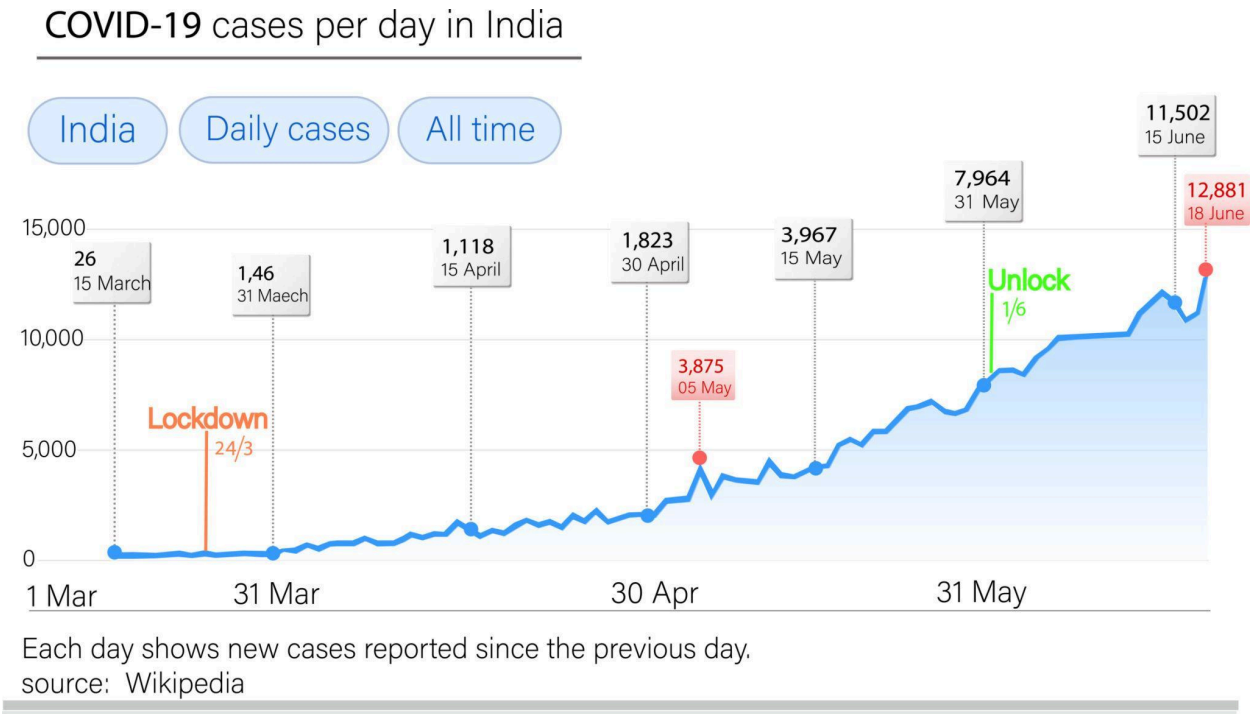
4. Donut Chart – Deaths vs Recovered

Shows proportion between recoveries and deaths.

Recovery dominates in all selected countries.

Useful for calculating survival/recovery rate visually.

5. Line Chart – Trend by Year & Quarter



- Tracks:
  - Confirmed cases
  - Active cases
  - Deaths
- Shows pandemic growth over time.
- Helps identify waves/peaks.

6. Horizontal Comparison Bars

- Compares Active vs Confirmed vs Recovered.
- Shows burden on healthcare vs recovery progress.

7. Slicers (Country/Region Filter)

- Interactive selection tool.
- Allows dashboard to update dynamically.
- Enables country-specific analysis (e.g., India, New Zealand, Cabo Verde).

Insights / Findings from Dashboard Analysis

1. Recovery Rate is Very High

- All dashboards show **recoveries > 90%** of total cases.
- Example:
  - India: ~95% recovered.
  - New Zealand & Cabo Verde also show dominant recovery share.

2. Death Rate is Comparatively Low

- Death percentage remains small relative to confirmed cases.
- Indicates improved treatment and healthcare response.

3. India Shows Highest Overall Impact

- Highest deaths (~1M).
- Highest confirmed (~41M).
- High active cases (~16M).
- Reflects large population + widespread transmission.

4. Smaller Countries Show Lower Fatalities

- Cabo Verde and New Zealand show deaths under a few thousand.
- Demonstrates better containment or smaller population exposure.

5. Pandemic Growth Trend

- Line graph shows steep rise from early 2020 to mid-2020.
- Confirms first major global wave period.

6. Active Cases Fluctuate

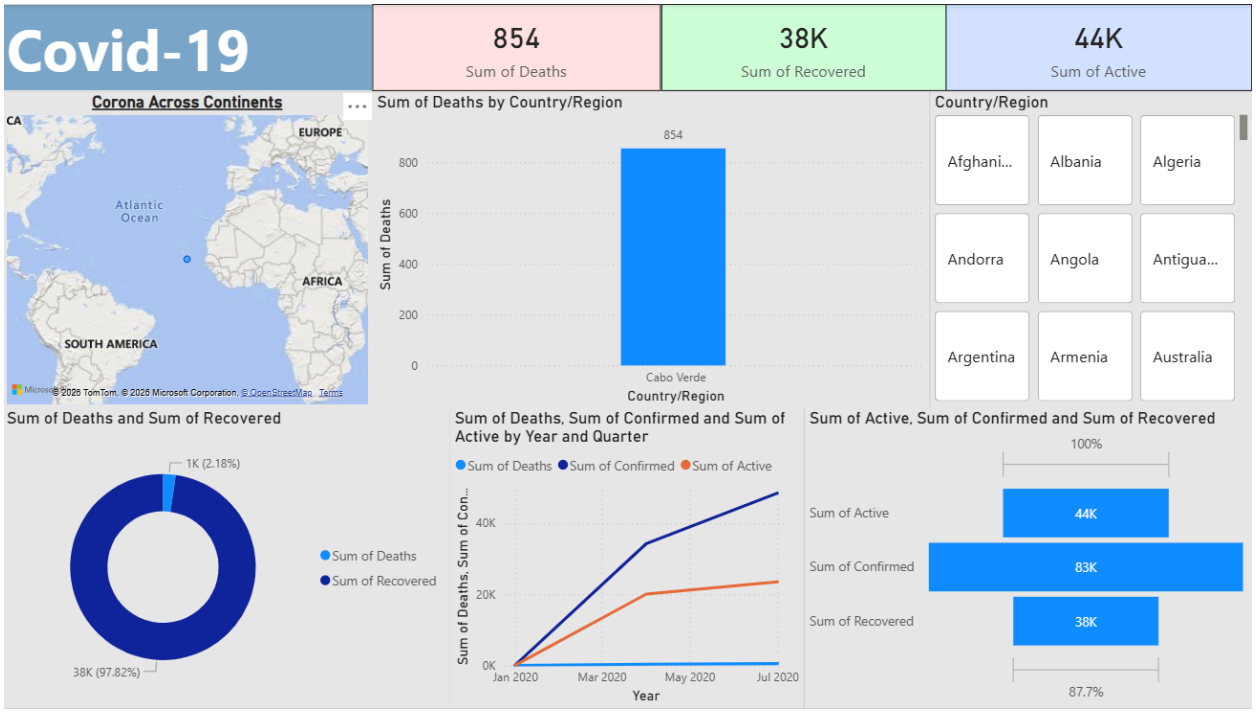
- Active cases rise with confirmed cases.
- Later decline indicates recovery acceleration.

7. Healthcare Burden Insight

- Countries with high active cases indicate strain on medical systems.
- India shows highest active load among selected regions.

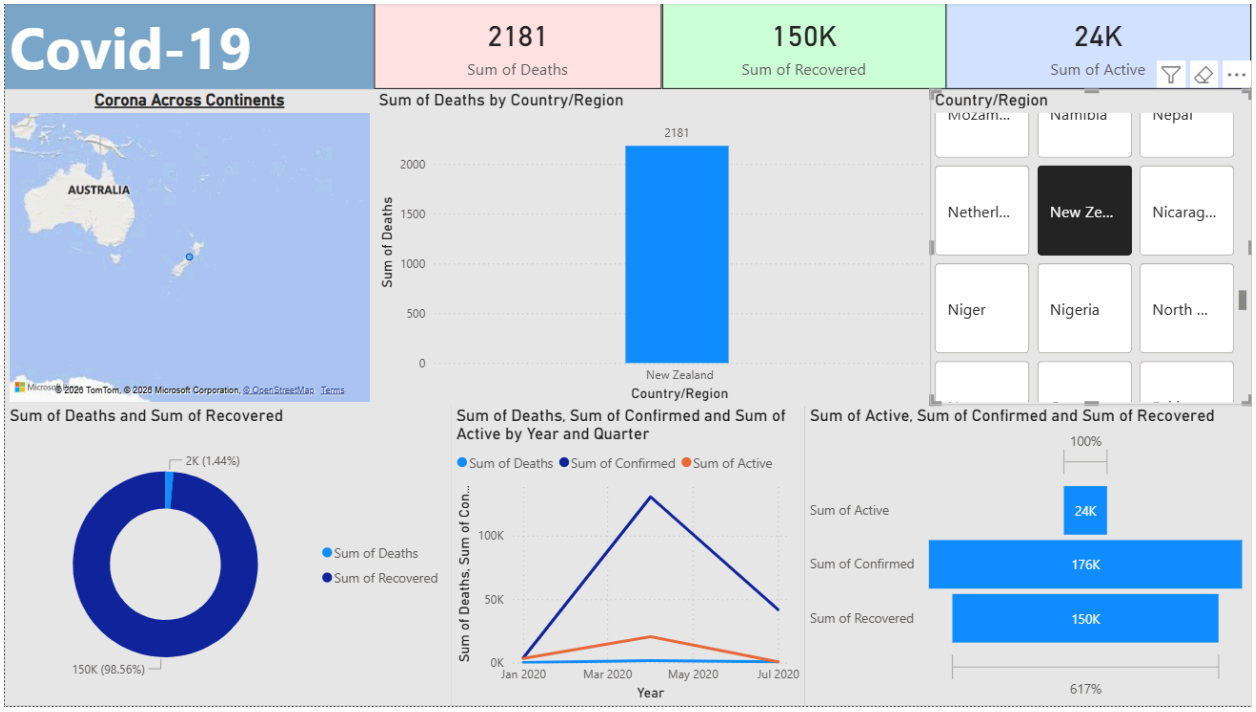
2. Screenshot

Overall:



For India:

For New Zealand:



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3. Conclusion

The COVID-19 dashboard provides a comprehensive visual analysis of pandemic data using multiple BI visualization techniques. By integrating KPI cards, geographic maps, bar charts, donut charts, and time-series graphs, the dashboard enables both macro and micro-level understanding of the outbreak.

From the analysis, it is evident that while COVID-19 spread extensively across continents, recovery rates remained significantly high in most regions. Countries with large populations, such as India, experienced the greatest number of confirmed, active, and fatal cases, highlighting the correlation between population density and virus transmission.

Smaller nations like New Zealand and Cabo Verde demonstrated relatively controlled fatality counts, suggesting effective containment strategies and healthcare management. The time-trend visualization further revealed the rapid surge of cases during early 2020, marking the initial pandemic wave.

Overall, the dashboard successfully transforms raw COVID-19 data into actionable insights, helping stakeholders monitor case progression, evaluate healthcare response, and support data-driven decision-making during global health crises.